

DOMAIN 04

AI / ML

D4-PS1

The CA in Your Pocket That Does Not Exist

BACKGROUND

India has 1.4 crore GST-registered MSMEs. A kirana store owner in Lucknow receives 500 to 2,000 invoices per month via WhatsApp images, printed digital invoices, machine-generated receipts, and scanned PDFs. They currently pay 15,000 to 40,000 rupees per month to a chartered accountant who manually enters this data into GST portals. The CA's real value is not data entry. It is knowing that a specific invoice carries the wrong HSN code and that fixing it will unlock 2,400 rupees in blocked Input Tax Credit this month.

THE PROBLEM

The GST portal offers no diagnostic tool that explains what is mismatched, why it matters, and what action to take in plain language. A small trader who tries to file independently misses ITC claims they are legally entitled to and loses real money every month. The core issue is an unstructured data paradox coupled with a language barrier. The reality of a small business is physical and messy while government compliance is digital and rigidly structured. Traders lack a bridge to translate their daily commercial artifacts into actionable compliance realities without paying a professional gatekeeper.

CONSTRAINTS

- GST portal backend APIs are not available to third-party developers.
- The target user has a smartphone but limited comfort with English interfaces.
- Output must be directly actionable by the trader without requiring professional interpretation.

Simulation Clause: *The trader's own GSTR-2B export, which any registered taxpayer can download directly from the portal as a PDF or spreadsheet, is a valid and available baseline. For prototyping purposes teams may generate their own clean dummy GSTR-2B datasets (CSV or Excel) rather than relying on actual government portal downloads.*

D4-PS2

The Right to Be Forgotten That Goes Nowhere

BACKGROUND

Indian professionals and students now routinely interact with multiple AI assistants across work and personal contexts. Each assistant collects interaction data, builds contextual memory, and retains user inputs under its own terms of service. India's Digital Personal Data Protection Act 2023 establishes a legal right to erasure of personal data. No technical infrastructure currently exists to exercise this right across AI tools simultaneously or to verify that erasure has occurred.

THE PROBLEM

A user who has interacted with multiple AI assistants over two years has no unified view of what each system retains about them, no way to understand what they originally consented to versus what is currently stored, and no mechanism to issue a single erasure or correction request that travels across all the platforms they have used. Each platform's consent record is siloed within its own interface, expressed in legal language most users cannot parse, and leaves no portable artifact in the user's possession. The right to erasure exists as a legal provision under the DPDP Act. It does not exist as a practical, exercisable capability for the ordinary user who has scattered their data across a fragmented AI ecosystem.

CONSTRAINTS

- No AI platform will grant API access to another platform's stored user memory or consent records.
- The DPDP Act's erasure provisions are legally enforceable but the technical enforcement infrastructure does not yet exist.
- The target user is a non-technical professional with no background in data law or platform architecture.
- Any solution must not require simultaneous cooperation from multiple platforms to function.

Simulation Clause: *Teams are not required to obtain real data from any AI platform's memory or consent systems. For the prototype, teams may generate mock interaction histories, simulated consent records, and synthetic user data across multiple fictional assistants.*